VIPER PILOT MILL

CASING REMOVAL AND SLOT RECOVERY MILL

MILLING

GENERAL DESCRIPTION

MILLING TOOL DESIGNED FOR THE EFFICIENT REMOVAL OF DOWNHOLE CASING STRINGS. IT IS USED MAINLY FOR OFFSHORE PLATFORM SLOT RECOVERY, WHERE INTERMEDIATE CASING IS REMOVED BACK TO THE SURFACE STRING OR CONDUCTOR SO THE WELL CAN BE RE-DRILLED TO A NEW TARGET INSTEAD OF BEING ABANDONED. WHERE THE CASING CEMENT IS POOR A MIX OF RETRIEVAL AND MILLING IS USED; WHERE CASING IS CEMENTED BACK TO THE SPOOLS, MILLING ALONE IS USED. THE BLADES CARRY A BRICK-PATTERN CARBIDE CUTTING STRUCTURE FOR HIGH PENETRATION RATE, IDEAL CUTTING SIZE AND LONG MILLING LIFE.

TECHNICAL CHARACTERISTICS

Tool series	Casing size	Body dia.	Blade dia.	No. of blades	Top pin conn.	Fishing neck length	Fishing neck dia.	Overall length	Wt (lb)
4500	4-1/2	4-3/4	5.250	3	3-1/2 IF BU	12	4-3/4	36	100
5000	5	4-3/4	5.813	3	3-1/2 IF BU	12	4-3/4	36	110
5500	5-1/2	4-3/4	6.300	3	3-1/2 IF BU	12	4-3/4	36	120
6000	6	4-3/4	6.875	3	3-1/2 IF BU	12	4-3/4	42	150
6600	6-5/8	5-3/4	7.640	3	4-1/2 IF BU	12	5-3/4	42	175
7000	7	5-3/4	7.906	3	4-1/2 Reg. BU	12	5-3/4	42	190
7600	7-5/8	6-1/4	8.750	5	4-1/2 IF BU	18	6-1/4	48	250
8600	8-5/8	6-3/4	9.875	5	5-1/2 Reg. BU	18	6-3/4	48	275
9600	9-5/8	8	10.875	5	6-5/8 Reg. BU	18	8	60	300
10700	10-3/4	9	12.000	5	7-5/8 Reg. BU	18	9	60	325
11700	11-3/4	10-1/4	13.000	5	8-5/8 Reg. BU	18	10-1/4	60	375
13300	13-3/8	11-1/2	14.625	5	8-5/8 Reg. BU	18	11-1/2	72	400
16000	16	14	17.250	5	8-5/8 Reg. BU	18	14	72	425

Dimensions in inches, weights in lb (approximate). Pilot stabilisation and guide or taper mill are dressed to the casing drift diameter. stabiliser sleeve available on casing sizes 7-5/8 in. and larger.



DESIGN FEATURES

- **Cutting structure:** carbide inserts in a brick pattern, developed to resist early wear and breakage.
- **Blades:** high-grade alloy steel, cutting edge set at the angle for maximum efficiency.
- **Extended blade length:** more footage per mill.
- **Stabiliser:** sleeve-type, changeable at the rig site; useful on eccentric casing strings.
- **Deviated holes:** blade design does not skin the next casing string.

OPERATING PARAMETERS

Weight on mill	5,000 - 10,000 lb
Cutting surface speed	250 - 340 ft/min
Rotary speed	From surface speed and casing O.D.

MILL STABILITY

- **Taper mill:** run directly ahead of the mill to clear junk or enlarge the casing stub.
- **Stabiliser:** same diameter as the mill, run immediately above it.
- **Sizing:** O.D. and stabilisation diameter set to avoid damage to outer casing strings.

HOW TO ORDER

Please specify: 1. tool series 2. size and weight of the casing to be milled 3. top pin connection.